

Mario Torres Danta

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Seville, Spain

EDUCATION

B.Sc. in Mathematics

2023 – 2027 (*Expected*)

University of Seville (US), Seville, Spain

Current GPA: 8.74 / 10.0 (Top 5% of cohort; based on 22 graded courses)

ACADEMIC HONORS AND DISTINCTIONS

Awarded "**Matrícula de Honor**" (**Highest Distinction**) in several courses, such as:

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| – General Topology (10.0) | – Numerical Analysis I (9.5) |
| – Measure Theory & Lebesgue Integration (10.0) | – Ordinary Differential Equations (9.4) |
| – Mathematical Optimization (9.2) | – Probability Theory (9.4) |

Additional "Sobresaliente" (A-Grade) in courses such as Local Geometry of Surfaces (10.0), Algebraic Structures, Complex Analysis, Linear Algebra II, Discrete Mathematics.

RESEARCH EXPERIENCE

Visiting Student & Undergraduate Researcher

Aug 2026 - Sep 2026

University of Copenhagen, Denmark

Project: Guided project on Mapping Class Groups, Teichmüller and Riemman Moduli Space, including Braid Theory and other related advanced topics.

Supervisor: Prof. Nathalie Wahl

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FUSRP Undergraduate Researcher

Jun 2026 - Jul 2026

The Fields Institute for Research in Mathematical Sciences, Toronto, Canada

Project: Guided project on Laurent symmetric functions, including their relation with representation theory.

Supervisor: Prof. Alistair Savage and Dr. Yaolong Shen

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Undergraduate Researcher (Departmental Collaborator) Sep 2025 - June 2026
Department of Geometry and Topology, University of Seville

Project: Advanced topics in Topology and Group Theory. A guided study complementing the standard undergraduate curriculum.
Supervisor: Prof. Ramón J. Flores

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Undergraduate Researcher (Departmental Collaborator) Sep 2024 - Jun 2025
Department of Algebra, University of Seville

Project: Classification of homographies and homologies in real ($\mathbb{P}^n(\mathbb{R})$) and complex ($\mathbb{P}^n(\mathbb{C})$) projective spaces.
Supervisor: Prof. Francisco Castro Jiménez

RESEARCH INTERESTS

Algebraic Topology (Homology, Cohomology, Homotopy Theory) · Abstract Algebra (Group Theory, Ring Theory) · Category Theory · Representation Theory

ACADEMIC ENGAGEMENT

Workshop on Geometric Topology Feb 2026
Informal Workshop
Role: Attendee. Participated as a listener in sessions focused on geometric topology and related structures. [[View Website](#)]

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ByMat Conference (Bringing Young Mathematicians Together) Nov 2025
University of Seville / RSME (Royal Spanish Mathematical Society)
Role: Attendee. Selected to participate in this conference for young researchers.
Highlight: Particular interest in plenary lectures by **Prof. Marithania Silvero** (Knot Theory) and **Prof. Alberto Rodríguez** (Differential Geometry/Topology), aligning with my research focus.

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Independent Advanced Study 2025 – Present
Proactively self-studying graduate-level texts to bridge the gap between undergraduate topology and current research:

- **B. Farb & D. Margalit, "A Primer on Mapping Class Groups":** Deepening understanding of surface homeomorphisms and Dehn twists (Pre-departure study for Copenhagen).
- **J. Munkres, "Topology" & "Algebraic Topology":** Mastery of fundamental groups and singular homology.
- **Sasho, "An Illustrated Introduction to Topology and Homology":** Visualization techniques for higher-dimensional structures.

SKILLS

- **Languages:** Spanish (Native), English (C1), French (B2)
- **Programming:** **Python** (Proficient; Certified by *Santander Open Academy*), **MathLab** (Academic use)
- **Technical:** **L^AT_EX** (Primary document preparation), **GitHub** (Repository management)

RECOMMENDATIONS

Prof. Ramón J. Flores

Professor of Geometry and Topology, University of Seville

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Web: <http://personal.us.es/ramonjflores>

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Prof. Alfonso Carriazo

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References (including research supervisors) and full academic transcript available upon request.